Are LEGO Themes Real Universes?
A Graph Analysis of the LEGO Catalogue

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Abstract

LEGO sells its sets under themes such as City, Star Wars, Technic or Friends. We ask whether these themes are separate "universes", each with its own parts, or labels put on a common pool of bricks, and whether LEGO has become more specialised since licensed themes appeared around 2000. From the Rebrickable database we build a bipartite graph between 9,062 sets (1990–2025) and the 28,856 parts they contain, and project it onto the sets: two sets are linked when the Jaccard similarity of their parts is at least 0.25. The resulting undirected, simple, weighted graph has 317,495 edges. We study its connectivity, distances and centralities, compare Louvain and label propagation communities with the official themes, measure the cohesion of each theme against a null model that shuffles parts between sets, and repeat the analysis decade by decade. Communities do not reproduce the themes (NMI 0.45): they mix themes of the same period. A few themes with their own building system (Duplo, Technic) are closed universes, while film franchises such as Harry Potter, Disney or Jurassic World keep only 3–12% of their links inside the theme; still, each of the 57 large themes is more cohesive than in all ten random graphs. The neighbours of a set guess its theme in 64% of cases. Finally, the share of theme-exclusive parts in a set halved between the 2000s and the 2010s (9.0% to 4.4%): sets grew larger with shared parts, so the specialisation hypothesis is not supported.

1 Introduction

LEGO organises its catalogue in themes. Some are built around a dedicated building system (Duplo bricks are twice the usual size, Technic uses beams and pins), others around a story or a licence (Star Wars since 1999, then Harry Potter, Marvel, Disney…). A fan's intuition is that each theme is a world of its own. But all themes are built from the same plastic bricks, and a 2×4 plate fits in a Star Wars ship as well as in a City police station. Whether themes are *universes* or *labels* is therefore a question about which parts sets share, which is naturally a network question [15].

Our objective is twofold. First, we test whether the official themes are visible in the structure of the catalogue: do sets of the same theme share more parts with each other than with other sets, and do communities found without any theme information coincide with the themes? Second, we test a hypothesis about the history of the catalogue: since the early 2000s and the multiplication of licences, sets would use more theme-specific parts, so that themes would be more separated in the graph. Bartneck and Moltchanova [2] showed on inventories from 1955 to 2015 that LEGO sets became larger and that the vocabulary of parts grew; we look at the same question from the point of view of themes.

The data do not come as a graph: we build a bipartite sets–parts graph and project it onto the sets (Section 2). Section 3 describes the measures and their parameters, Section 4 the results, and Section 5 the answers and the limits. The code, the data and the instructions to reproduce every number are in the GitHub repository `https://github.com/enzontl/lego-universes`.

2 Dataset Presentation

2.1 Source

Rebrickable [20] is a community database of the LEGO catalogue, used by fans to find which sets they can build with the parts they own. It publishes its whole catalogue as CSV files. We use six tables: `themes` (theme tree), `sets` (number, name, year, theme, number of parts), `inventories` and `inventory_parts` (which parts, in which colour and quantity, each set contains), `parts` and `part_categories`. The files were downloaded on 6 November 2026 and are included in the repository; since Rebrickable updates them continuously, numbers may change slightly with a later download. Rebrickable's terms allow the catalogue data to be used for any purpose and ask users to credit the source. We chose this dataset

first because we like LEGO and wanted to look at the bricks we grew up with in a new way, and also because it is complete (28,567 sets), well documented, and because the relation between sets and parts is a genuine network that has, to our knowledge, rarely been analysed from the angle of themes.

2.2 From tables to a graph

Selection. We keep sets released from 1990 to 2025 with at least 50 parts, so that very small sets (polybags, key chains) do not dominate. We remove themes that are not building sets (Gear, Books, Key Chain, Bulk Bricks, Service Packs, Supplemental, Collectible Minifigures). Each set is attached to its *root theme*, the top of Rebrickable's theme tree (for example *Star Wars* for *Star Wars / Ultimate Collector Series*). This leaves 9,062 sets in 127 root themes.

Bipartite graph. A set is linked to each part it contains. A part is identified by its shape (`part_num`) regardless of colour, because a red and a blue 2×4 brick are the same mould. Spare parts are ignored, only the first version of each inventory is used, and minifigures are not counted (their parts are stored in separate inventories). The bipartite graph has 9,062 sets, 28,856 parts and 735,437 set–part edges.

Projection. Two sets A and B, with sets of distinct parts P_A and P_B, are linked with the Jaccard weight [12]

$$w(A, B) = \frac{|P_A \cap P_B|}{|P_A \cup P_B|} \in [0, 1], \tag{1}$$

and the edge is kept only if $w(A, B) \geq 0.25$. A raw count of shared parts would link every large set to almost every other set; Jaccard compares the overlap with the size of the two sets. The threshold is needed because almost all pairs of sets share at least one basic part: at 0.15 the graph already has 4.6 million edges and 97% of the sets in one component. Figure 1 shows how the graph changes with the threshold. We keep 0.25, which keeps three quarters of the sets in one component while removing the weakest links, and we present as conclusions only the results that hold at other thresholds.

Graph type. Sharing parts is symmetric, so the graph is *undirected*. There is at most one edge per pair of sets and no loop, so it is a *simple* graph, and it is *weighted* by w. Nodes carry the set number, name, year, theme, root theme and number of parts.

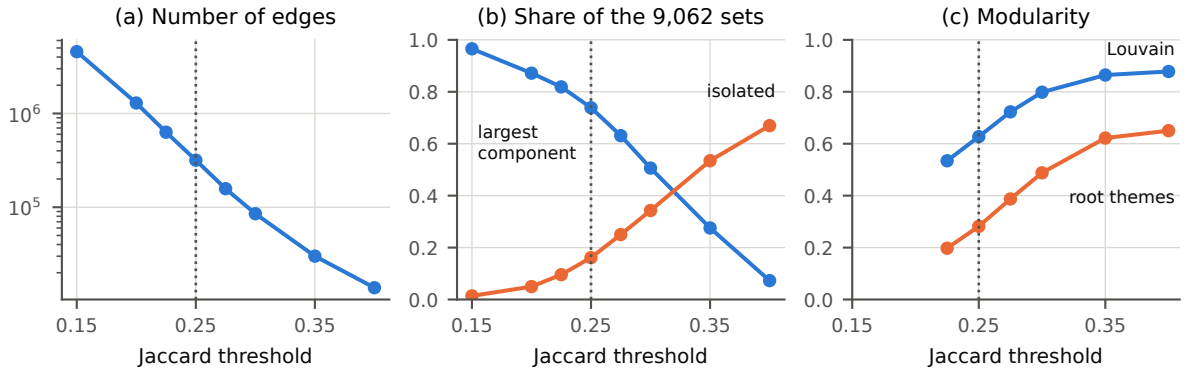


Figure 1: Effect of the Jaccard threshold (dotted line: the chosen value 0.25). (a) Number of edges, log scale. (b) Share of sets in the largest component and isolated. (c) Modularity of the Louvain partition and of the root themes; Louvain was not run below 0.225 (graph too dense).

2.3 Graph statistics

Table 1 summarises the projected graph. It is sparse (density 0.008) but each set has on average 70 neighbours. The degree distribution is heavy-tailed but bends down on a log–log scale and stops near 1,000 (Figure 2a), so it is not a power law. On the part side (Figure 2b), 77% of the parts appear in a single root theme, while 51 parts appear in 100 themes or more (the 1×2 plate is in 73% of the sets).

Table 1: Statistics of the projected set graph (threshold 0.25). *On the largest component.

Nodes (sets)	9,062	Connected	no
Edges	317,495	Components	1,640
Density	0.0077	Isolated sets	1,462 (16%)
Mean degree	70.1	Largest component	6,694 sets (74%)
Maximum degree	997	Second component	282 sets (Duplo)
Median edge weight	0.27	Diameter / radius*	16 / 8
Mean weighted clustering	0.14	Mean distance*	≈ 3.8

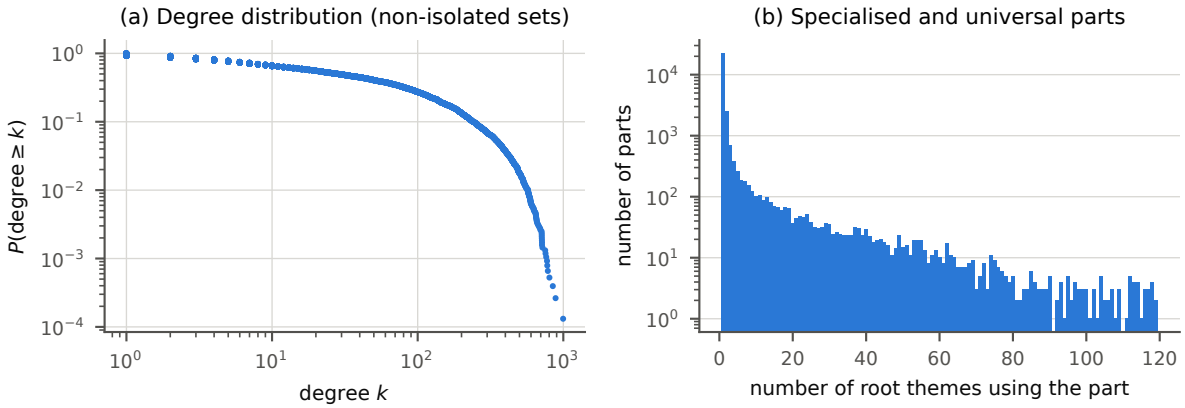


Figure 2: (a) Complementary cumulative degree distribution of the set graph. (b) Number of parts according to the number of root themes in which they appear (bipartite graph, log scale).

3 Description of the Analysis

Software. The analysis is written in Python 3.11 with NetworkX 3.7 [10], pandas, NumPy, SciPy, scikit-learn [18] and matplotlib. The projection is computed with sparse matrices. Exact package versions are frozen in `uv.lock`, every random step uses the seed 42, and the README of the repository lists the commands that produce each table and figure; the figures of this report are drawn from the result files by `report/make_figures.py`. The Gephi images (Figure 4) were made with Gephi [3].

Connectivity and distances. We compute the connected components. Since the graph is not connected, the diameter and the radius are computed on the largest component only, with the exact bounding algorithm of NetworkX (`usebounds=True`) [22]. Distances count edges (unweighted): a path is a chain of sets that pairwise share at least a quarter of their parts. The mean distance is estimated from 200 random sources. The weighted clustering coefficient follows Onnela et al. [17].

Centrality. We use four measures, each answering a different question about a set. The *degree* counts the sets that share at least a quarter of their parts with it, and the *strength* sums the weights of these links: both tell how "generic" a set's parts are. The *betweenness* [6] counts how often a set lies on shortest paths between other sets; it identifies sets that bridge otherwise distant regions of the catalogue. Since a strong similarity should mean a short distance, the length of an edge is $1/w$ [14]; the measure is estimated from 500 random sources [7]. The weighted *eigenvector* centrality [5], computed on the largest component, is high for sets connected to other well-connected sets, i.e. the core of the graph.

Communities. We run the weighted Louvain algorithm [4] (resolution 1, seed 42), which maximises the modularity [16]

$$Q = \frac{1}{2m} \sum_{i,j} \left(w_{ij} - \frac{s_i s_j}{2m} \right) \delta(c_i, c_j), \tag{2}$$

where s_i is the strength of set i and $2m$ the sum of all strengths. We compare the partition found with the root themes in two ways: the modularity of the theme partition itself, against two baselines (themes shuffled at random among sets with the same sizes, and a partition by decade), and the normalised mutual information (NMI) [9] and adjusted Rand index (ARI) [11] between communities and themes.

The 1,462 isolated sets each form a one-set community, which inflates the NMI, so we also report NMI and ARI without them. To check that the conclusion does not depend on the algorithm, we rerun Louvain at resolutions 0.5, 2 and 4 and run label propagation [19], in its asynchronous weighted version and in its deterministic semi-synchronous version [8]. The modularity of all these partitions is evaluated at resolution 1 so that the values are comparable.

Cohesion of a theme. For a theme t, let W_t be the total weight of the edges inside t and $\mathrm{vol}(t)$ the sum of the strengths of its sets. The *cohesion* (internal share)

$$c_t = \frac{2W_t}{\mathrm{vol}(t)} \tag{3}$$

is the share of the theme's link weight that stays inside the theme: 1 for a closed universe, close to 0 for a label. If links ignored themes, the expected value would be $\mathrm{vol}(t)/2m$. We report it for the 57 root themes with at least 30 sets.

Null model. A theme can look cohesive only because its sets are large or contain frequent parts. To control for this, we shuffle the bipartite sets × parts matrix with the curveball algorithm [21]: it repeatedly picks two sets and redistributes the parts they do not share. Every set keeps exactly its number of parts and every part its number of sets; only the link between parts and themes is destroyed. After 20 rounds, 22% of the set–part pairs are still in place. We draw 10 null matrices (seeds 42 to 51) and project each of them. At the threshold 0.25 a null graph has almost no edges (117 to 235), so we keep its strongest edges, as many as in the real graph (cut-off ≈ 0.16), and compute the same cohesion. With 10 draws the smallest possible empirical p-value is $1/11 \approx 0.09$, so we report how many draws reach the real value and the ratio between the real cohesion and its mean in the null graphs.

Theme graph and theme prediction. We contract the set graph into a graph of the 127 root themes: the weight of the edge between themes a and b is the total weight W_{ab} of the links between their sets, and its *lift* $W_{ab}/(\mathrm{vol}(a)\,\mathrm{vol}(b)/2m)$ compares it with the weight expected if links ignored themes (lift > 1: more linked than chance). We also ask whether a set's neighbours reveal its theme: each neighbour votes for its own root theme with the weight of the edge, the set's own theme is never used, and isolated sets get no prediction. We compare the success rate with always predicting the most frequent theme and with drawing a theme at random according to the theme frequencies.

Evolution over time. We rebuild the graph separately for 1990–1999, 2000–2009, 2010–2019 and 2020–2025 and recompute the modularity of the themes, Louvain and the NMI. Specialisation is measured on the bipartite graph, independently of the threshold: for a set, the *exclusive share* is the share of its distinct parts that appear in no other root theme of the 1990–2025 catalogue; we also report the number of exclusive parts per set and the number of distinct parts per set. Because a sticker sheet is a part number specific to one set, we repeat the analysis without the Rebrickable category "Stickers", and at the thresholds 0.25 and 0.35.

Reused code and AI tools. No existing code analysing the Rebrickable data was reused. Graph algorithms (components, diameter, centralities, Louvain, label propagation, modularity) come from NetworkX, and NMI and ARI from scikit-learn; the curveball algorithm was implemented by us from its description in [21]. We used Claude Code (Anthropic), an AI coding assistant, to help write and test the analysis code, to search for related studies, and to draft and format this report. All numbers in the report are produced by the scripts of the repository, and we checked the modelling choices and the interpretations ourselves.

4 Results

4.1 Connectivity, distances and central sets

The graph is not connected (Table 1). The largest component holds 74% of the sets; 16% of the sets share at least a quarter of their parts with no other set and are isolated. The second component (282 sets) is almost entirely Duplo (216 sets) and Duplo-based educational sets: Duplo bricks are not compatible in size with the rest of the catalogue, and the graph shows it as a separate island. Smaller components are

also single themes: LEGO Brand Store (23 sets), Clikits (21), Belville (12). Inside the largest component, any two sets are at most 16 steps apart and about 3.8 steps on average. The eight sets of the center (eccentricity 8) are large models for adults (Ford Mustang, Volkswagen T2 Camper Van, Land Rover Defender, Voltron, Tantive IV); the three sets of the periphery are atypical (a Botanicals flower bouquet, an educational kit, a 2007 Racers car). The center of the catalogue is therefore not a theme but a type of set: large, recent, and built from common parts.

Table 2: Five most central sets for each measure (root theme, year).

Degree	The Brick Moulding Machine (LEGO Exclusive, 2021) 997; R2-D2 (Star Wars, 2021) 885; Luke Skywalker's Landspeeder (Star Wars, 2022) 847; Old Trafford – Manchester United (Creator, 2020) 803; The Dark Falcon (Star Wars, 2024) 783
Betweenness	Architecture Studio (Architecture, 2013) 0.037; Fallingwater (Architecture, 2009) 0.030; NANO QUEST Challenge Kit (FIRST LEGO League, 2006) 0.024; Amusement Park (Factory, 2005) 0.018; Skyline (Factory, 2005) 0.016
Eigenvector	The Dark Falcon (Star Wars, 2024) 0.070; Ghost & Phantom II (Star Wars, 2023) 0.070; R2-D2 (Star Wars, 2021) 0.070; Luke Skywalker's Landspeeder (Star Wars, 2022) 0.070; Jazz Quartet (LEGO Ideas, 2022) 0.068

The centralities (Table 2) tell the same story. The sets with the highest degree and strength (the order is almost the same for both) are large display models from 2019–2024, mostly Star Wars, which use a wide range of standard parts and thus resemble hundreds of other sets. Eigenvector centrality concentrates on the same group: recent large Star Wars, Ideas, Marvel and DC sets form the densest core of the graph. Betweenness highlights a different kind of set: bridges whose inventory mixes part families. Architecture Studio is a box of white and transparent bricks of all kinds, the FIRST LEGO League kits combine Technic and classic bricks, and the 2005 Factory sets were designed from fan models with unusual part assortments. A theme like Star Wars is central because it is large and generic, not because it is a hub of specific parts.

4.2 Communities are not themes

Table 3: Partitions of the set graph compared with the root themes. Modularity is always evaluated at resolution 1; NMI and ARI exclude the 1,462 isolated sets. "≥ 10": number of groups with at least 10 sets; "largest": share of the sets in the largest group.

Partition	≥ 10	largest	modularity	NMI	ARI
Root themes (127)	87	8%	0.282	1	1
Root themes shuffled at random	87	8%	0.000	–	–
Decades (4)	4	38%	0.360	–	–
Louvain, resolution 0.5	24	22%	0.610	0.43	0.10
Louvain, resolution 1	35	19%	0.627	0.45	0.10
Louvain, resolution 2	39	12%	0.588	0.49	0.15
Louvain, resolution 4	59	7%	0.477	0.54	0.22
Label propagation, asynchronous	32	27%	0.580	0.49	0.09
Label propagation, semi-synchronous	36	28%	0.577	0.47	0.08

Louvain finds 1,677 communities, of which 1,462 are isolated sets and only 35 have at least 10 sets, with a modularity of 0.63 (Table 3). The theme partition reaches only 0.28. This is far above shuffled themes (≈ 0), so themes do structure the graph, but much less than the best partition. The agreement between communities and themes is moderate (NMI 0.45 without isolated sets) and the ARI is low (0.10). Figure 3 shows why. The four largest communities, which together hold more than half of the sets, are mixtures of themes from the same period: community C1 (median year 2004) mixes Town, Star Wars, Creator and Space, C3 (2016) mixes City, Star Wars, Friends and Ninjago, and C2 (2023) Friends, Star Wars, Marvel and City. Only the themes with their own building system form their own community: Technic (C4, 66% Technic, with Bionicle and Racers, which use Technic parts) and Duplo (C6, 81% Duplo). The Gephi drawing of the graph (Figure 4) gives the same picture: the colour patches of the communities cut across the colours of the themes.

At the threshold 0.25, the partition by decade even has a higher modularity than the themes (0.36 against 0.28). This comparison, however, depends on the threshold: at 0.30 the themes are ahead (0.49 against 0.43) and at 0.35 clearly so (0.62 against 0.48), because a higher threshold keeps only very similar sets, which tend to belong to the same theme (these two values are computed by `report/extra_checks.py`). We therefore do not claim that time structures the catalogue more than themes; we only conclude that large communities group sets by period rather than by theme.

The conclusion does not depend on the algorithm. Label propagation gives partitions close to Louvain (NMI 0.47–0.49, modularity 0.58). Increasing the Louvain resolution raises the NMI from 0.43 to 0.54 and the ARI from 0.10 to 0.22: at resolution 4 the giant mixed communities break up (the largest goes from 22% to 7% of the sets) and the granularity becomes close to that of the themes. Even then, the ARI stays at 0.22, far from a match.

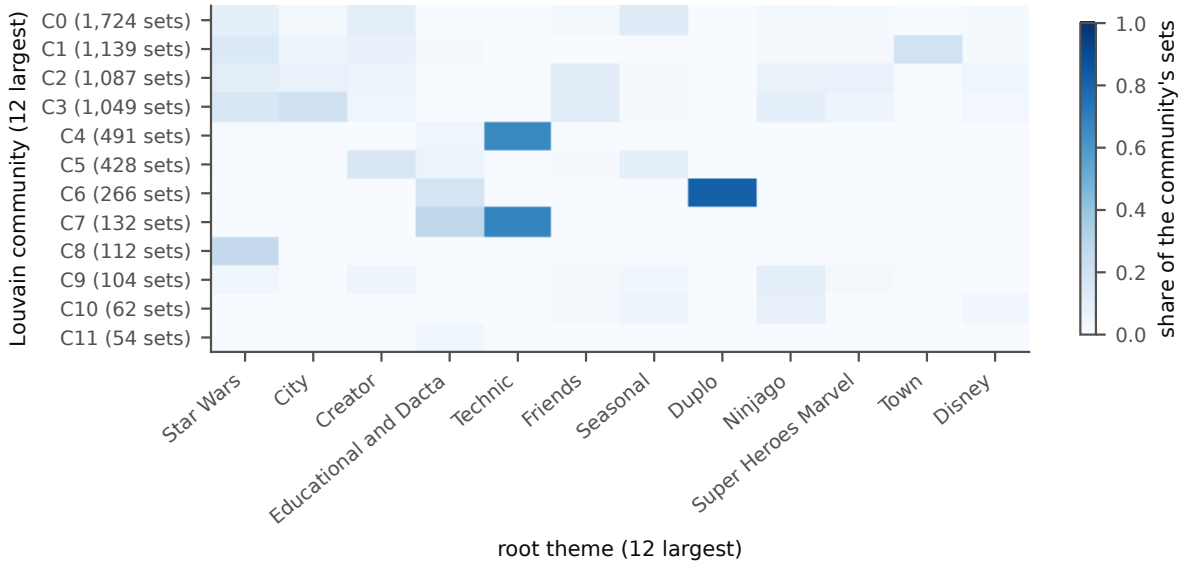


Figure 3: Composition of the 12 largest Louvain communities in terms of the 12 largest root themes (share of the community's sets; columns do not sum to 1 because other themes are omitted).

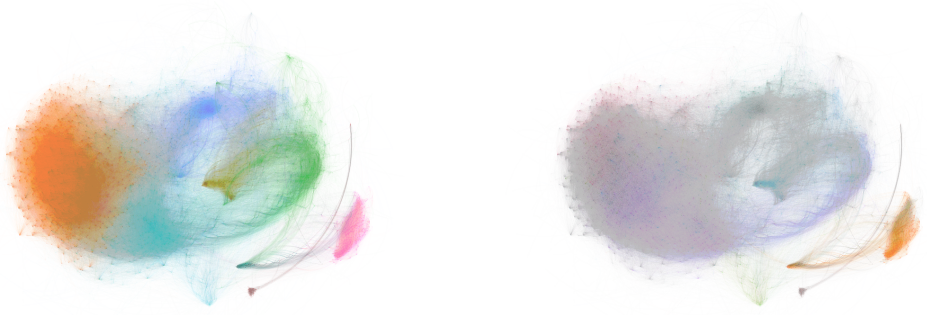


Figure 4: The set graph drawn with Gephi (ForceAtlas2 [13], isolated sets removed, same layout in both panels). Left: coloured by Louvain community; right: coloured by root theme. Colours come from Gephi's partition palette, in which many small themes get similar colours: the figure is meant for the overall comparison, not to identify single themes.

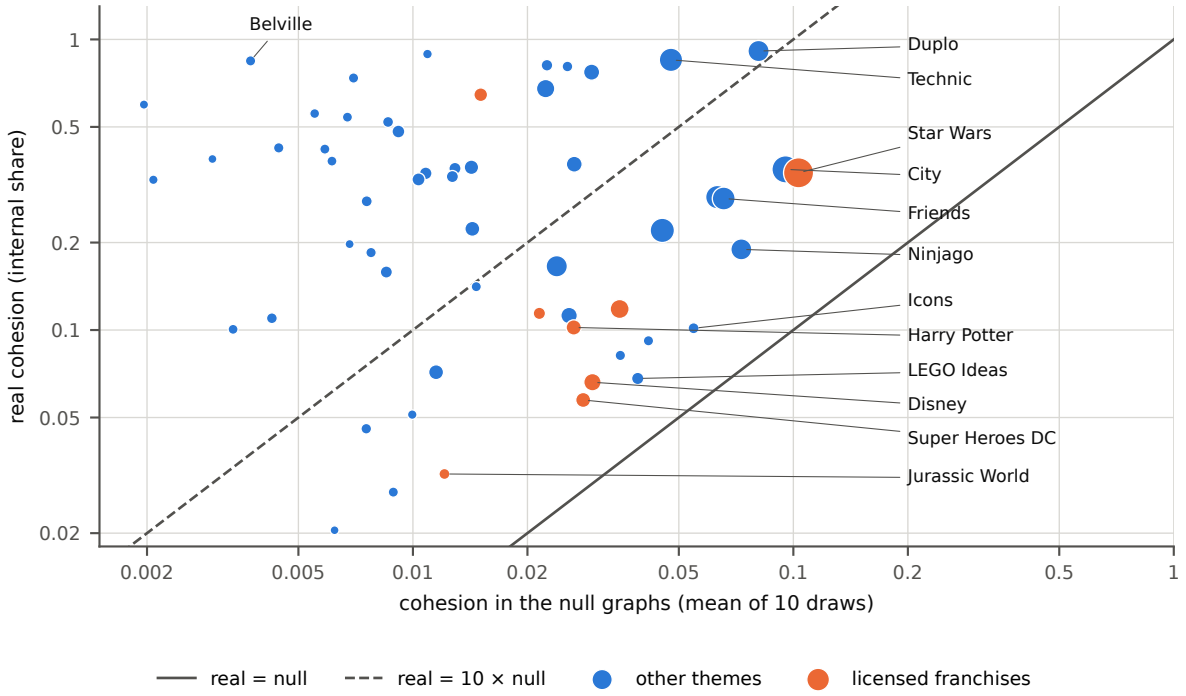


Figure 5: Cohesion of the 57 root themes with at least 30 sets, against their mean cohesion in 10 null graphs (log scales; point size: number of sets). All themes lie above the line "real = null"; most licensed franchises lie between the two lines, i.e. less than 10 times above chance.

Table 4: Selected root themes: number of sets, cohesion c_t, mean cohesion in the null graphs and ratio, and theme prediction by neighbour vote (recall: share of the theme's linked sets correctly guessed; precision: share of the predictions of this theme that are correct).

Root theme	sets	cohesion	null	ratio	recall	precision
Duplo	339	0.91	0.081	11.3	0.88	0.82
Technic	419	0.85	0.048	17.8	1.00	0.75
Brickheadz	181	0.77	0.029	26.2	1.00	0.46
Town	246	0.68	0.022	30.3	0.93	0.82
Castle	102	0.48	0.009	52.6	0.92	0.92
City	582	0.36	0.095	3.7	0.76	0.90
Star Wars	721	0.35	0.103	3.4	0.86	0.41
Friends	402	0.28	0.066	4.3	0.83	0.62
Creator	457	0.22	0.045	4.9	0.57	0.39
Ninjago	331	0.19	0.073	2.6	0.59	0.76
Super Heroes Marvel	266	0.12	0.035	3.4	0.40	0.66
Harry Potter	159	0.10	0.026	3.9	0.45	0.76
Disney	216	0.07	0.030	2.2	0.33	0.82
Super Heroes DC	152	0.06	0.028	2.1	0.11	0.50
Jurassic World	65	0.03	0.012	2.6	0.17	0.67
LEGO Ideas	92	0.07	0.039	1.7	0.00	0.00

4.3 Which themes are universes?

The cohesion of the 57 themes with at least 30 sets ranges from 0.03 to 0.91 (median 0.29; Table 4). At the top are themes with a dedicated building system or a strong visual style: Duplo (0.91), Clikits (0.89), Technic (0.85), Belville (0.84), Mixels and DOTS (0.81), Brickheadz (0.77). At the bottom are licensed franchises and catch-all themes: Harry Potter keeps only 10% of its link weight inside the theme, Disney 7%, Super Heroes DC 6%, Jurassic World 3%.

The null model qualifies the reading of these low values (Figure 5). When parts are shuffled between sets, the modularity of the themes falls from 0.28 to 0.004–0.005, and *all 57 themes are more cohesive than in each of the 10 null graphs*. The median ratio to the null model is 19, and 33 themes are at least 10 times above chance; those with their own parts are far above it (Castle 53, Town 30, Technic 18). Film franchises are only 2 to 4 times above chance (Harry Potter 3.9, Marvel 3.4, Jurassic World 2.6, Disney 2.2, DC 2.1), and the theme closest to chance, LEGO Ideas (1.7), is by design a collection of fan projects with no common world. Minecraft is the exception among licences (cohesion 0.65, 43 times chance), because its sets are built from dedicated block-shaped parts. The ratio is lower for large themes, whose null cohesion is high simply because they hold many sets, which is why Star Wars and City (3.4 and 3.7) are close to the film franchises in ratio while their cohesion is three to five times higher. Film franchises are therefore not mere labels, but *weak universes*: their sets share a little more than chance, much less than Technic or Duplo.

The theme graph (Figure 6) shows how themes relate. It has 127 nodes and 2,487 edges (density 0.31); 7 small themes (2 to 19 sets) have no link at all. The heaviest edges go through the two largest generic themes, Creator–Star Wars (6,942 links between sets) and Star Wars–Icons, but their lift is close to 1 (1.10 and 1.38): they are heavy only because both themes are large. The lift reveals families instead: Castle–Pirates (56), Pirates–Adventurers (45), Train–Town (20) and Bionicle–Hero Factory (15) are themes of the same period or the same part system; Hero Factory replaced Bionicle in 2010.

The prediction experiment confirms the picture. Voting among neighbours guesses the root theme of 63.9% of the 7,600 linked sets (61.6% without weights; 53.6% if isolated sets count as failures), against 8.4% by always predicting Star Wars and 3.1% at random. Averaged over the 57 large themes, the success rate is 59%. Recall is highest for the closed universes (Technic and Brickheadz 100%, Town 93%, Castle 92%, Duplo 88%) and lowest for the licensed and catch-all themes (DC 11%, Jurassic World 17%, LEGO Ideas 0%), whose sets are mostly predicted as Star Wars. Recall alone can be misleading: Star Wars is guessed for 86% of its sets, but it is predicted for 1,337 sets of which only 41% are Star Wars. As the largest theme with generic parts, it attracts the vote of its neighbours (most of its false predictions are Creator, Marvel, Ninjago and DC sets). Brickheadz shows the same effect (precision 0.46): it absorbs Seasonal and Promotional sets built in the same style.

4.4 Did LEGO become more specialised?

Table 5: Evolution by period. Modularity of the root themes at thresholds 0.25 and 0.35; exclusive parts with stickers kept (removed in brackets). Exclusive parts do not depend on the threshold.

			modularity		exclusive parts		parts
Period	sets	themes	0.25	0.35	share	per set	per set
1990–1999	927	30	0.52	0.54	8.9% (8.3%)	4.6 (4.2)	57
2000–2009	1,780	57	0.31	0.47	9.0% (8.4%)	4.2 (3.9)	61
2010–2019	3,411	70	0.32	0.67	4.4% (3.9%)	2.9 (2.4)	79
2020–2025	2,944	56	0.17	0.62	4.4% (4.0%)	3.8 (3.3)	103

Our hypothesis was that sets became more theme-specific after 2000. The data do not support it (Table 5, Figure 7). The share of a set's parts that no other theme uses stays at 9% in the 1990s and 2000s, then halves to 4.4% in the 2010s and 2020s. This result does not depend on the threshold, which only affects the projection, and it is not caused by stickers: without them the share goes from 8.3% to 4.0%. It also holds under other modelling choices (minimum set size of 20 or 100 parts, parts distinguished by colour, no excluded theme, parts present in more than 20% of the sets removed): in every case the share in the 2010s and 2020s (3.8–6.4%) is about half of the share in the 1990s (8.5–11.7%).

The counts explain this decline. The number of exclusive parts per set falls less (4.6 in the 1990s, 2.9 in the 2010s) and rises again in the 2020s (3.8), while sets grow from 57 to 103 distinct parts on average. Sets have become larger, as Bartneck and Moltchanova found [2], but the parts they added are mostly

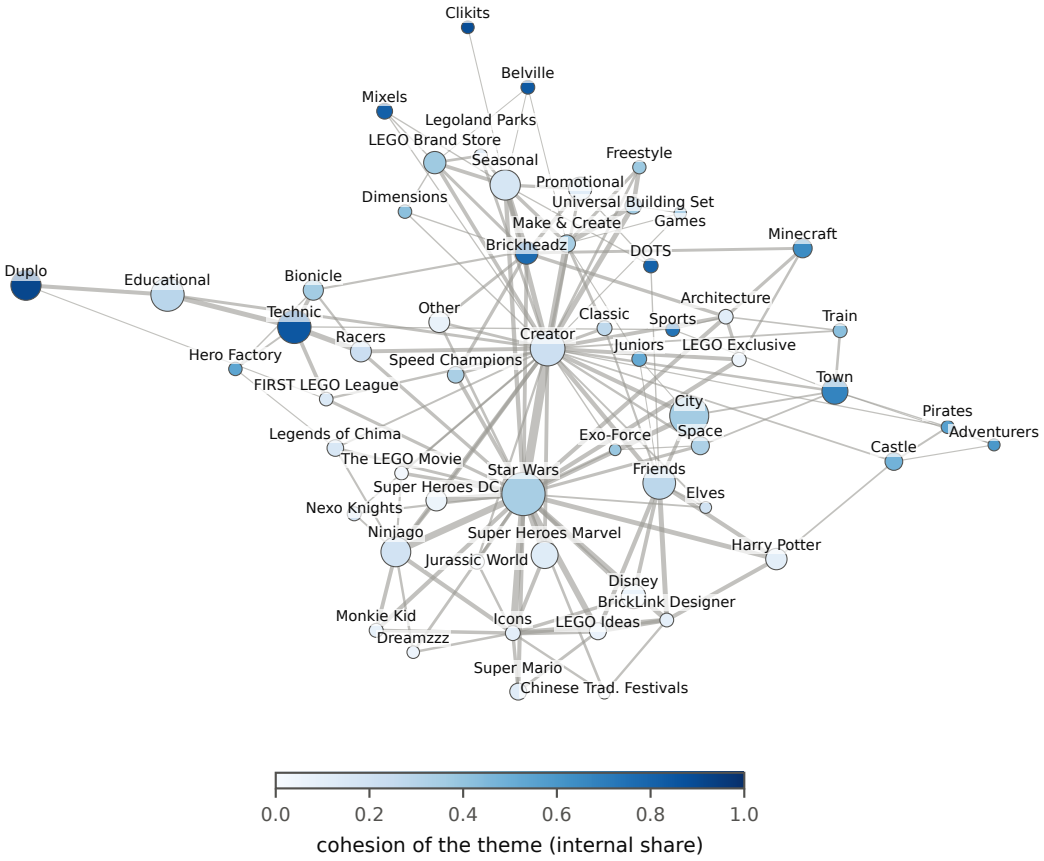


Figure 6: Theme graph: one node per root theme with at least 30 sets (size: number of sets; colour: cohesion c_t), with the 3 heaviest edges of each theme (width: total Jaccard weight between the two themes).

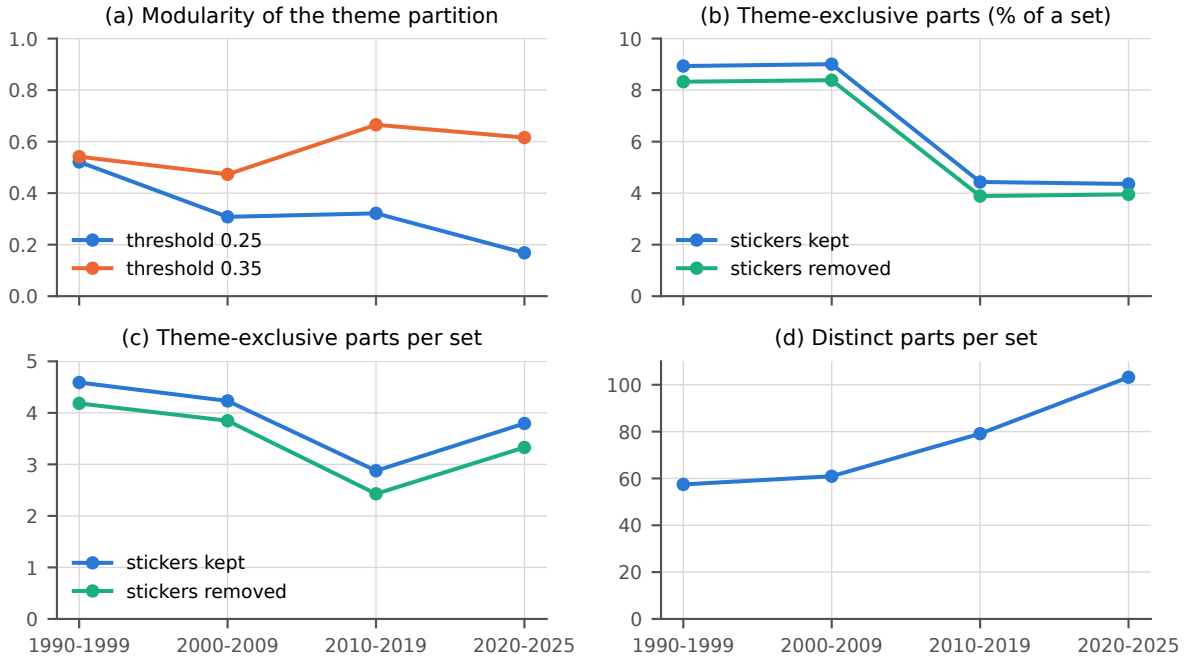


Figure 7: Evolution by period. (a) Modularity of the root themes at two thresholds. (b) Share and (c) number of a set's parts that appear in no other root theme, with and without stickers. (d) Mean number of distinct parts per set.

parts shared across themes. Their conclusion that sets became more specialised is about the growing vocabulary of the catalogue; at the level of themes, we find the opposite: a smaller share of each set is specific to its theme.

The structural indicators are less conclusive. At the threshold 0.25, the modularity of the themes falls from 0.52 to 0.17 and the NMI between Louvain communities and themes from 0.70 to 0.42, which would suggest that themes are melting into each other. But at 0.35 the modularity of the themes rises instead (0.54, 0.47, 0.67, 0.62). Recent sets are larger and share more standard parts, so at a fixed threshold they get many more links (39 links per set in the 2010s, 79 in the 2020s, against 24 in the 1990s), and the modularity at 0.25 partly measures this density. We therefore do not report a trend in modularity, only the robust decline of theme-exclusive parts.

5 Conclusions

LEGO themes are, for the most part, not separate universes. Sets mainly link to sets of the same period, and the communities of the graph mix themes of the same era (NMI 0.45 with the themes, ARI 0.10, whatever the algorithm). Only a few themes are closed worlds, those with their own building system or style: Duplo, Technic, Clikits, Belville, Brickheadz. Film franchises such as Harry Potter, Marvel, DC, Disney and Jurassic World are weak universes: only 3 to 12% of their link weight stays inside the theme, but they remain 2 to 4 times more cohesive than chance, and all 57 large themes beat all ten null graphs. Over time, LEGO did not become more theme-specific: the share of theme-exclusive parts in a set halved between the 2000s and the 2010s, because sets grew with parts shared across the catalogue. Centrality measures point in the same direction: the most central sets are large, recent, generic models, and the bridges are sets that mix part families.

Limitations. The projection depends on the threshold. We only draw conclusions that hold from 0.25 to 0.35; the comparison between themes and decades, and the trend of modularity over time, do not, and are not claimed. A printed part or a sticker sheet has its own part number, which inflates exclusive parts; removing stickers changes little, but printed parts are still counted separately. Rebrickable is maintained by volunteers, and we use only the first version of each inventory and ignore minifigures. The null model uses 10 draws, so p-values cannot go below 0.09, and it is compared at equal density rather than at the same threshold. The projection discards part of the bipartite structure [1]. Finally, Louvain and asynchronous label propagation depend on the seed and, for near-ties, on the machine: on a second computer, with the same code and versions, the NMI of 2000–2009 was 0.52 instead of 0.53.

Future work. Community detection directly on the bipartite graph [1] would avoid the threshold altogether. Weighting parts by quantity or by colour, adding minifigures, and studying the graph of parts (which parts are used together) would refine the notion of a theme's own parts. More null draws would give proper p-values.

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